California Housing
Price Prediction

Regression Analysis · SGDRegressor vs Random Forest

MACHINE LEARNING PROJECT

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Objective

Predict median house prices across California census districts using machine learning regression models trained on demographic and geographic features.

$207,300

Average Median House Value in Dataset

RMSE

$49,823

R^2 Score

0.83

Dataset

1990 California Census · 20,640 districts · 9 columns

Pipeline

Load → Clean → Encode → Scale → Train → Evaluate

Models

SGDRegressor (L2) and Random Forest Regressor

Goal

Minimize RMSE · Maximize R^2 on held-out test set

Features Overview

Feature	Type	Description
longitude	Float	Geographic coordinate (west = more negative)
latitude	Float	Geographic coordinate (north = larger)
housing_median_age	Int	Median age of houses in district
total_rooms	Int	Total rooms per district
total_bedrooms	Int	Total bedrooms (207 nulls → imputed)
population	Int	Number of people in district
households	Int	Number of household units
median_income	Float	Median income (in tens of thousands)
ocean_proximity	Category	Distance from ocean (5 categories)

Target Variable

median_house_value

Median house price for
all houses in a district

⚠ 207 nulls in total_bedrooms
→ Filled with column mean

20,640

rows in dataset

03 / DATA PREPROCESSING

1

Load Data

Read CSV · 20,640 rows
10 features + target

2

Null Imputation

total_bedrooms:
207 nulls → mean fill

3

Feature Eng.

Ratio features:
rooms_per_household

4

One-Hot Encode

ocean_proximity →
5 binary columns

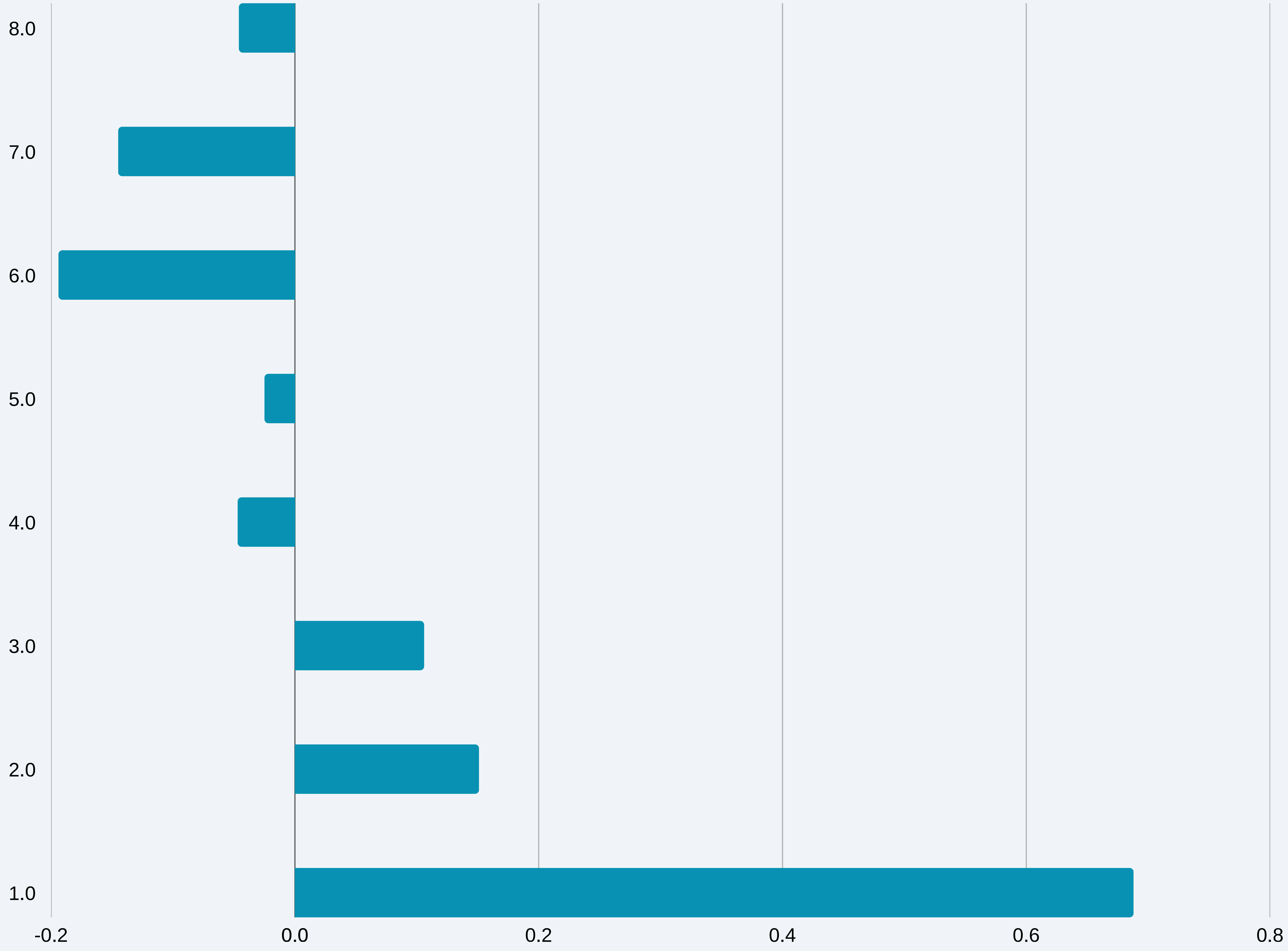
5

Robust Scale

RobustScaler on
numeric features

Train / Test Split: 80% training (16,512 rows) · 20% test (4,128 rows) · random_state = 42

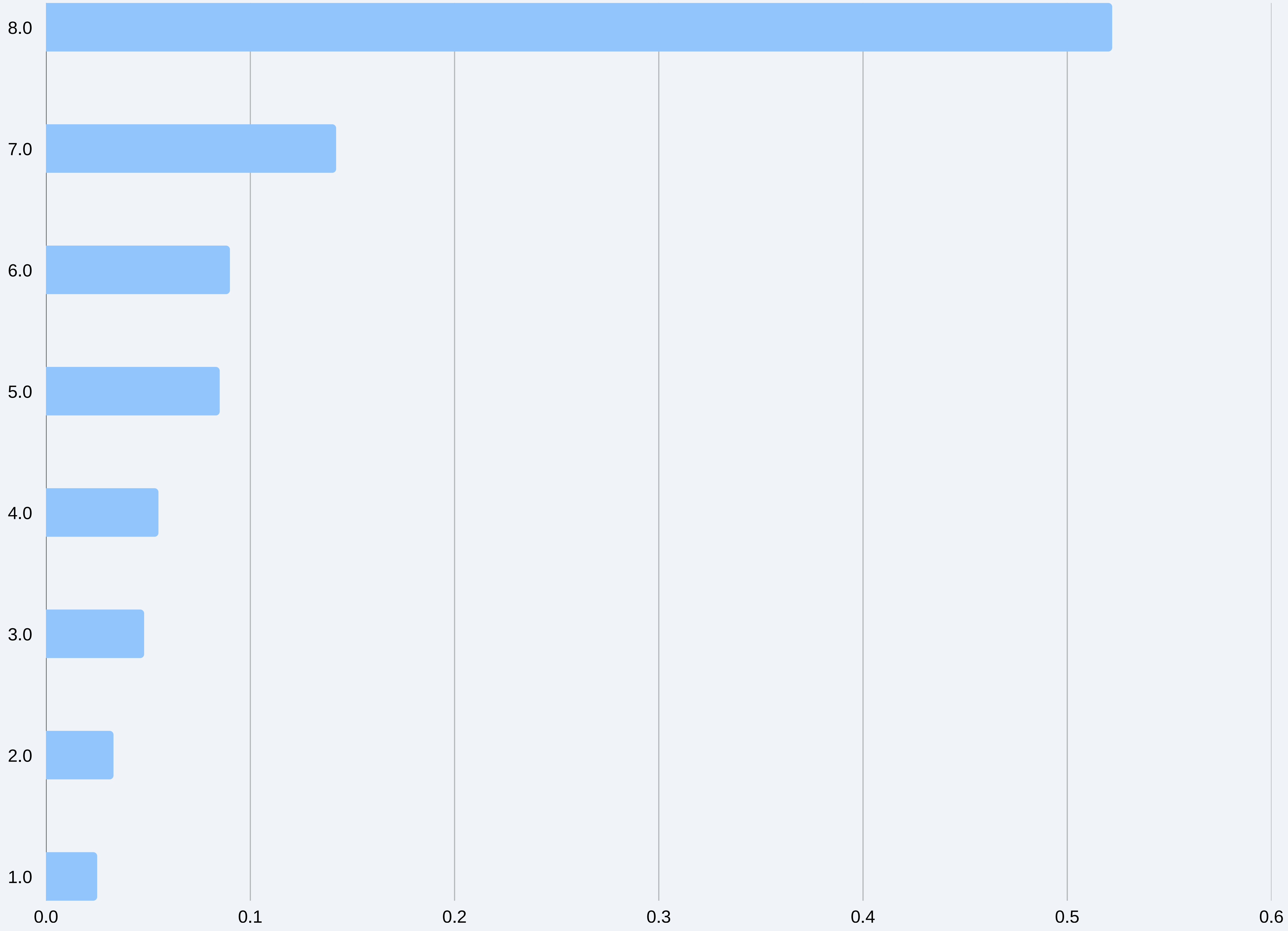
Correlation with Target (median_house_value)



Key Findings

- MedInc has strongest positive correlation (0.688) — higher income = higher price
- AveOccup negatively correlated: overcrowded districts have lower property values
- Geographic features (Lat/Lon) show significant correlation — location matters
- Price distribution is right-skewed with a hard cap at $500,001 in raw data

05 / FEATURE IMPORTANCE (Random Forest)



52.2%

MedInc is the dominant feature

AveOccup: 14.2%

2nd place: overcrowding penalty

Geo features: 17.5%

Lat + Lon combined influence

Top 3 features explain >84% of model

Configuration

Regularization	**L2 (Ridge)**
Alpha	**0.001**
Learning Rate	**adaptive**
Loss	**squared_error**
Max Iterations	**1000**
Scaling	**RobustScaler**

RMSE

$84,512

R^2 Score

0.61

How SGDRegressor Works

- Stochastic Gradient Descent updates weights one sample at a time
- L2 regularization (Ridge) penalizes large coefficients to reduce overfitting
- Adaptive learning rate adjusts step size during training
- RobustScaler preprocessing is critical — SGD is sensitive to feature scale
- Limitation: assumes linear relationship between features and target

Configuration

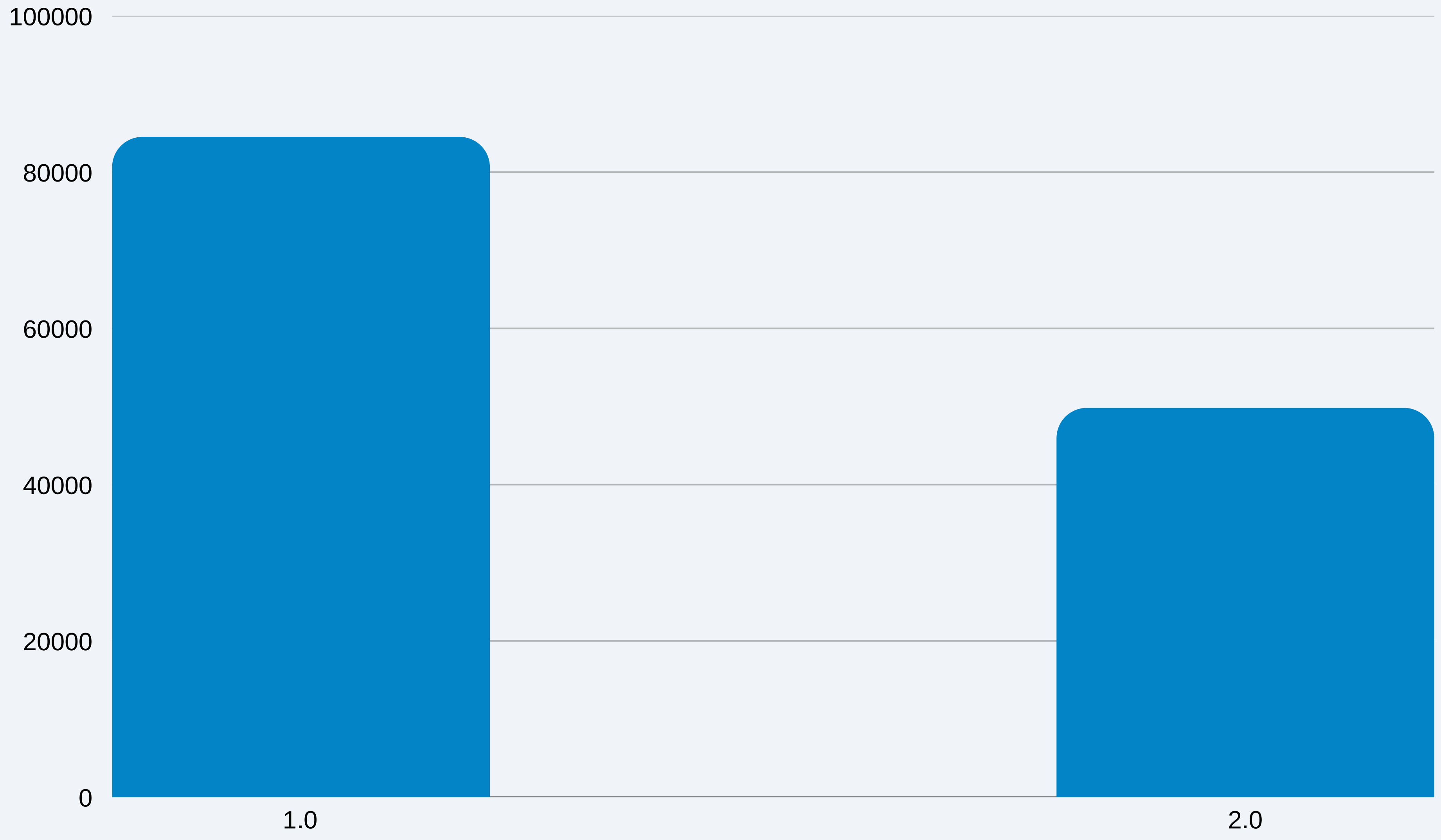
n_estimators	**100 trees**
max_features	**auto (√n_features)**
max_depth	**None (fully grown)**
bootstrap	**True**
min_samples_leaf	**1**
Scaling needed	**No**

RMSE	R^2 Score
$49,823	**0.83**

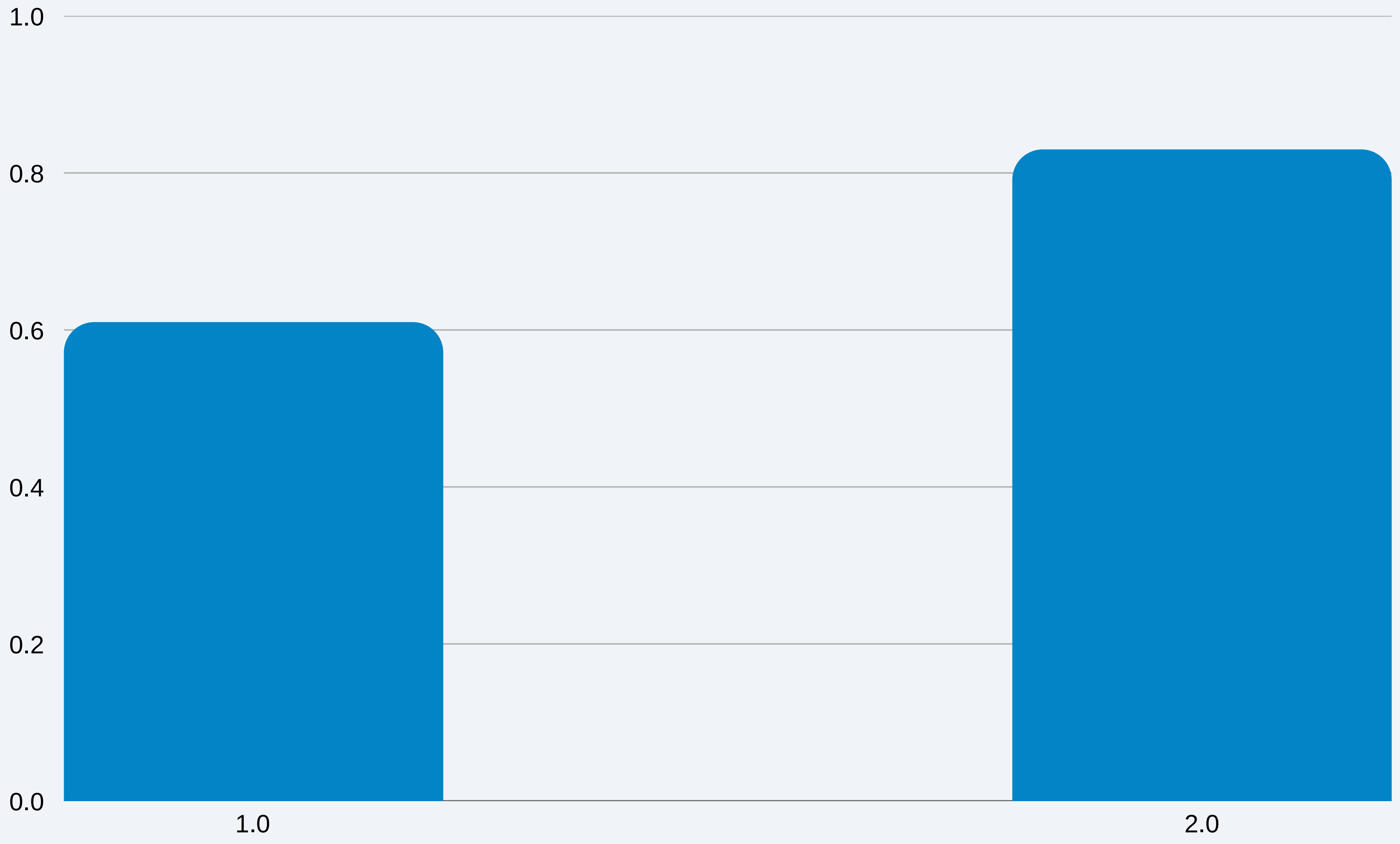
How Random Forest Works

- Builds 100 decision trees on random subsets of data (bagging)
- Each tree sees a random subset of features → reduces correlation
- Final prediction = average of all 100 tree predictions
- Handles non-linear relationships automatically without feature scaling
- Provides feature_importances_ for model interpretability

RMSE — Lower is Better



R^2 Score — Higher is Better



Metric	SGDRegressor	Random Forest	Winner
RMSE	$84,512	$49,823	RF ↓41%
R^2 Score	**0.61**	0.83	RF ↑36%
Scaling	Required	Not needed	RF

08 / RESULTS & EVALUATION

What does R^2 = 0.83 mean?

The Random Forest model explains 83% of the variance in house prices. This is a strong result for a first model without hyperparameter tuning.

SGD R^2=0.61 — 61%

RF R^2=0.83 — 83%

RMSE $49,823 ≈ 24% of avg price ($207K)
Good baseline — room for improvement with tuning

Next Steps to Improve

1. Hyperparameter tuning via GridSearchCV or RandomizedSearchCV
2. Try Gradient Boosting (XGBoost, LightGBM) for higher accuracy
3. Cross-validation for more robust performance estimates
4. Feature engineering: price_per_room, log-transform target
5. Neural network approach for complex non-linear patterns

CONCLUSION

Key Takeaways

- ✓ Random Forest outperforms SGDRegressor: RMSE $49,823 vs $84,512 (41% improvement)
- ✓ Median income (MedInc) is the strongest predictor — 52% of feature importance
- ✓ Geographic features (Latitude + Longitude) together contribute 17.5% to predictions
- ✓ Proper preprocessing (imputation, one-hot encoding, scaling) is critical for model quality
- → Future work: XGBoost + hyperparameter tuning could push R^2 beyond 0.88

Random Forest	0.83	$49,823	+41%
Best Model	R^2 Score	RMSE	Improvement